Infobits

The Microbial Type Culture Collection and Gene Bank (MTCC), a national facility established in 1986 is funded jointly by the Department of Biotechnology (DBT) and the Council of Scientific and Industrial Research (CSIR), Government of India. The MTCC, housed at the Institute of Microbial Technology (IMTECH), Chandigarh, has established itself as a distinguished culture collection centre for microbial resources in India. It is an affiliate member of the World Federation for Culture Collections (WFCC) and is registered with the World Data Centre for Microorganisms (WDCM). The main objectives of this national facility are to act as a depository, to supply authentic microbial cultures and to provide related services to the scientists working in research institutions, universities and industries.

8.3 Whittaker's System of Classification

It is the five kingdom classification. In the 20th century, advances in cell biology and interest in evolutionary biology led scientists to question the two or three-kingdom classification schemes. In 1969, Robert H. Whittaker proposed a system which recognizes five kingdoms of living things: Monera (Bacteria), Protista, Fungi, Plantae and Animalia (Table 8.1).

Whittaker's system of classification is based on 1) complexity of cell structure 2) mode of nutrition 3) body organization 4) phylogenetic or evolutionary relationship.

Monera: This kingdom includes all prokaryotic organisms. Unicellular microorganism such as Mycoplasma, Bacteria, Actinomycetes and Cyanobacteria are grouped under kingdom Monera.



Phylogeny is the evolutionary history of organisms that refer to the relationship between life forms.

Table 8.1 Properties of Whittaker's five kingdoms

Kingdom	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell organization	unicellular	unicellular	Multicellular and unicellular	Multicellular	Multicellular
Cell Wall	Present in most	Present in some absent in others	Present	Present	Absent
Nutritional Class	Phototrophic, heterotrophicor chemoautotrophic	Heterotrophic and phototrophic	Heterotrophic	Phototrophic	Heterotrophic
Mode of nutrition	Absorptive	Absorptive or ingestive	Absorptive	Mostly Absorptive	Mostly ingestive

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